Nestify user guide

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Getting to know Nestify

Everything happens in one window. The numbers on the picture match the list below it.



#	Part	What it is for

1	**Project name**	Click it for the project menu: new house, open, save, import a plan or a 3D model, export, project details and Rescue.
2	**Open and Export**	Open a saved house; export plans, pictures, DXF and the 3D model.
3	**View tabs**	3D, Floor Plan, Elevation and Section: four ways to look at the same house.
4	**Undo, Redo, Save**	Ctrl+Z, Ctrl+Y and Ctrl+S.
5	**Top buttons**	Online library, Nestify Studio, camera, photo-real picture, cost, building rules, tips, settings, help and about (see The top bar).
6	**Tool rail**	Every tool, in groups (Select and draw, Change, Build ...). Hover over a button to see what it does; the Tool reference topics explain each one. Drag its right edge for 1 to 4 buttons across.
7	**Project Structure**	Everything in the house - floors, walls, roof, slab, rooms, objects, scenes - in a list. Click to select; the eye hides, the lock locks.
8	**Floors and view chips**	Switch floor or add one (+ Floor), show room labels, saved Scenes, Views, Sides for the plans, and Tags.
9	**3D view**	Draw and change the house here. Drag to turn round, right-drag to slide, wheel to zoom.
10	**Axes**	Which way X (red), Y (green) and Z (up, blue) point as you turn the view.
11	**Properties and Styles**	The settings of what is selected: sizes, material, colour, doors and windows; Styles changes how the whole view looks.
12	**Tips card**	What the tool you picked does and what to do next. Drag it anywhere; the bulb button hides it.
13	**Hand and Find a tool**	The hand slides the view. Type what you want to do in Find a tool (Ctrl+F) and pick it.
14	**Mouse hints**	What a click and a drag do right now.
15	**Status line**	Messages: what just happened and what to do next.
16	**Snap and Grid**	Snap to corners, walls and guides; show the grid.
17	**Running cost**	The cost of the house as it stands, updated after every change. Click for the full bill.
18	**Size box**	While drawing or moving, type a length or size here (or just type) and press Enter.
19	**Units**	mm, cm, m, inches or feet for everything you see and type.
20	**Fold the side panels**	The arrows (or [and]) fold the side panels away for a bigger view.

Moving round the view: drag with the left mouse button to turn round the house, drag with the right button (or the arrow keys, or the hand) to slide, and roll the wheel to zoom towards the mouse. F frames the whole building, T looks straight down.

Every tool works the same way: pick it on the rail (or press its key, or find it with Ctrl+F), read the Tips card, click in the view, type a size when you want an exact one, and press Enter to finish or Esc to stop. Ctrl+Z undoes anything.

Quick start: your first house

Eight steps from nothing to plans and a cost. Each step has its own topic further on.

1. **Start a house.** Press New house (top left) and describe it in your own words ("3 bedrooms, 2 bathrooms, double garage"), pick a ready house, import a plan picture, or draw it yourself.
2. **Draw or change the walls.** Wall (L): click corner after corner, type lengths for exact walls. Select a wall to change its length, height, thickness and material on the right.
3. **Doors and windows.** Door/window: pick one from the pictures and click on a wall. Select it to change its size, style and which way it opens.
4. **The roof.** The roof follows the plan by itself. Select it to change its style, pitch, overhang and covering, or open the Roof designer.
5. **Furnish and finish.** Furnish fills every room; the Library adds more. Paint (B) puts colours and materials on anything.
6. **The site.** Site makes the plot slope; Path and Road, Retaining wall, Dig and Hills shape the ground round the house.
7. **Check the cost and the rules.** The running cost is at the bottom; the calculator button opens the full bill. The clipboard button checks the building rules.
8. **Hand it over.** Export the plans (PDF, DXF), the council pack, photo-real pictures, a walk video or the AR page for phones.

The top bar

The buttons along the top of the window, from left to right.

	Button	What it does
	Project name	Click the name (Untitled at first) for the project menu: a new house (describe it, a ready plan, or draw it), open, save, import a plan or a 3D model, export, project details (owner, erf, address) and Rescue.
	Open	Open a saved house (Ctrl+O).
	Export	Plans as PDF, DXF and pictures, the council pack, the 3D model, the AR page and walk videos.
	3D	The 3D model, where you draw and change things.

	Floor Plan	The plan from above, drawn like a printed plan with sizes, doors and room names.
	Elevation	The house seen straight from the front, back and sides.
	Section	The house cut open, to look inside and print sections.
	Undo	Ctrl+Z: takes back the last change (as many as you like).
	Redo	Ctrl+Y: puts it back again.
	Save	Ctrl+S: saves the house.
	Online library	Free furniture, doors, windows, plants and materials for your plan.
	Nestify Studio	Design the selected thing freely (the roof, a shape, furniture, a wall).
	Camera	Camera angles: front, sides, top, corners, eye level, or your own.
	Photo-real	A photo-real picture of this view.
	Quantities and cost	Bricks, cement, sand, concrete, roof, doors, windows, finishes and site works, with prices you can change.
	Building rules	Checks the house against the building rules (SANS 10400 rules of thumb) and lists what passes and what needs attention.
	Tips	Shows or hides the Tips card.
	Settings	Nestify's settings.
	Help	How to use Nestify, the welcome tour and every tool explained.
	About	About Nestify and its developer.

Start a new house

The first time Nestify opens, a short welcome shows the four steps (get a plan, shape it, check it, hand it over). Open it again from Help (?) → Welcome tour.

Press **New house** (the house-plus icon, top left):

- **Describe it** in your own words, for example "3 bedrooms, 2 bathrooms with en-suite, open plan, double garage and a study, about 180 m2" or "double storey, 4 bedrooms, laundry". The line underneath shows what Nestify understood as you type. Press **Build it** and you get a full house: walls, passage, doors, windows sized for daylight, room names, a garage with driveway and car, stairs for two storeys, and furniture, lights, sockets and plumbing in every room. Every generated plan passes the building rules check.

- **Ready houses**: Modern villa, Modern family home and Double-volume home (see Ready modern houses).
- **Draw it yourself**: on a flat 2D plan, or straight in 3D.
- **From a plan you have** (photo, scan or PDF): let Nestify trace it into walls, or lay it in 3D and draw over it.
- **Empty 3D page** to draw your own with the wall, rectangle and line tools.

Everything it makes is ordinary walls and objects, so you edit it like any other plan.

Ready modern houses

The start card (house-plus icon, top left) has four ready houses to start from, each one built wall by wall like a house you drew, with rooms named, furniture in the rooms and lights in the ceilings:

- **Hillside villa** - three storeys with a two-storey wing: two hip roofs at different heights in orange tiles with white boxed eaves, cars parked under the house, a glass entrance hall two storeys high, glass balconies, curved flared stairs from the driveway up to the front door, a paved driveway, a boundary wall, a pergola on the lawn and palms.
- **Modern villa** - two storeys: a glass living room that opens to the pool, glass running into the corner with no post, stone and timber cladding, a pivot front door, a glass balcony upstairs, stepped flat roofs with deep dark overhangs and warm light under them, a timber double garage, a slatted privacy screen at the pool, a boundary wall with a gate and a numbered pillar.
- **Modern family home** - one storey: a low dark metal roof, white walls with timber panels, tall glass sliding doors, a pivot front door under its own porch roof, aluminium fins hiding the yard, a double garage, paving and stepping stones.
- **Double-volume home** - two storeys: a living room two storeys high (the floor above it is open, with a glass gallery round it), corner glass up both floors, a pivot door in a stone wall under a floating concrete canopy, timber slats in front of the main bedroom, a flat roof with a deep overhang.

Everything in them can be changed, moved or deleted, and Ctrl+Z brings back what you had before.

Modern houses: Make modern and the MODERN tools

The **MODERN** group on the left rail turns any house into a modern one, or adds the pieces one by one:

Button	What it does
Make modern	One click on the house you drew: a flat dark roof with a deep overhang, white walls, slim black window frames, low windows made tall, sliding doors made glass sliders, timber on the garage wall, a stone wall at the front door with a timber pivot door and a concrete canopy over it. Ctrl+Z puts it all back.
Feature wall	Click two opposite corners on a wall's face and pick stone, timber, face brick, off-shutter concrete, charcoal or white render: just that area changes. Doors and windows under it stay open.

Canopy	Click two corners on the plan, then pick the height (over a door, at the top of the walls, a carport, or your own): a floating 200 mm concrete slab.
Parapet	A 600 mm wall round the top of the house that hides the flat roof behind it.
Screen	Click where it starts and where it ends: vertical timber slats, charcoal aluminium fins or horizontal louvres. The whole screen is one group: it moves and deletes as one, and shows as one line in Project Structure.
Corner glass	Click a corner where two walls meet, then pick the glass: frameless (only glass, clear, grey or bronze tinted), a glass wall with a black, charcoal, bronze or white frame, or slim frame windows. It runs into the corner from both sides, with no post, and stops before any door or window already there.
Double volume	Click a room on the lower floor: the floor above it opens up, so the room is two storeys high. Put a glass railing round the opening upstairs.
Lights	Downlights in every room, warm LED strips under the eaves and canopies, uplights at the foot of feature walls. Pressing it again replaces them. Turn Lights on, or take an evening photo, to see them.

The **Pivot** front door (a wide door turning on a pin, with a long bar handle) is on the door list too.

Tool reference: Select and draw

Picking things and drawing the plan: walls, rooms, shapes, openings and guides.

	Tool	What it does and how to use it
	**Select** `Space`	Click a wall, door, window, roof, room or object to select it; its settings show on the right. Ctrl+click adds more to the selection. Drag the round grips to change it, drag empty space to turn the view. Esc always comes back here.
	**Wall** `L`	Click where the wall starts, then corner after corner; double click or Esc to stop. Type a length while drawing (for example 4.2 and Enter) for an exact wall. Walls snap to each other, to guides and to right angles, and rooms are found by themselves when walls close a space.
	**Line** `N`	Click point after point to draw any outline on the ground, on a top or on the side of any shape or wall (A switches to an arc for the next side). Close it on the first point, then Push/Pull it up into a solid.
	**Arc**	The Line tool starting with a curved side: click the start, the end, then how far it bulges.
	**Curved wall**	Click where the wall starts, then where it ends, move the mouse to bend it and click. It is one smooth wall: doors and windows put in it follow the curve. Use Curve on it later to change the radius, flip it or straighten it. Or click on a box or any shape where the curve starts, then where it ends, and bend it: only the part between your two dots curves.

	**Rectangle** R	Click one corner, then the opposite corner (or type the size, like 4,3) - on the ground, on top of a shape or on top of a wall - or on the SIDE of a shape or wall, then push it in or pull it out as a new piece. Then pull it up into a box. A box can be used as a building: cut doors and windows into it.
	**Room**	Click two corners: four walls are drawn round the space, joined at the corners, and the room is named and measured.
	**Trace a room**	With a plan picture on the ground, click inside a room on the picture: walls are drawn round it along the lines of the picture.
	**Circle** C	Click the middle, then the edge (or type the radius). Pull it up into a smooth cylinder - a round tank, column or round room.
	**Polygon**	Pick how many sides (3 to 12), click the middle and a corner; pull it up into a prism.
	**Shape push/pull** U	On any face, draw a rectangle, circle, polygon or your own dots (or Auto: the whole face, sized to meet the next wall), then push that area in (a recess, a niche) or pull it out (a bay, a ledge). Drawn on a wall's top and pulled up, it becomes a new shape of its own on the wall.
	**Opening** D	Drag along a wall where the opening goes: then pick door, window or passage. The width is the length you dragged.
	**Guides**	Click where the guide line goes, then V for vertical or H for horizontal. Walls and shapes snap to guides; use as many as you like. Esc stops.
	**Protractor**	Click the corner, then a point on each arm: the angle shows, and a guide line can be left at that angle.
	**Plan picture**	Choose a photo, scan or PDF of a plan. It lies on the ground; click two points whose distance you know and type it to scale it, then draw the walls over it (or use Trace room).
	**Door or window**	Pick a door or window from the pictures, then click on any wall or on the side of a box. In a box it gets a frame (jambs, head and sill). Click again for more; Esc stops.

Tool reference: Change

Moving, turning, resizing, cutting, copying and tidying what is drawn.

	Tool	What it does and how to use it
	**Move** M	Select something, then drag it to where it goes. Type a distance while moving for an exact move.
	**Rotate** Q	Select an object and press Rotate: a glass protractor appears round it. Point at the face to turn about (top: it turns round; a side: it tips; the front: it rolls) and drag round the circle - it clicks every 15 degrees. Walls turn about their middle; -15 / +15 step them.

	**Push/Pull** `P`	Click a face of a shape (or a wall's side) and drag it in or out; type a distance for an exact push or pull. A part of a face pulled out becomes a new piece of its own. An outline you just drew with Shape push/pull waits for it: pick Push/Pull and drag.
	**Scale**	Type a percentage (110 makes it 10% bigger) for what is selected.
	**Split** `S`	Draw a line across walls or any shape: they are cut in two along it, each piece its own.
	**Offset**	Click the top of a shape you drew and type a distance: a new outline inside (minus) or outside it, ready to Push/Pull into a border, a step or a kerb.
	**Sweep**	Pick a profile (round pipe, square bar, skirting, moulding, kerb or gutter) and its size, then click the path it runs along; Enter finishes.
	**Array copies**	Many copies of what is selected: in a row (type the step across, along and up, like 1.2, 0) or round a circle (for example p 45 for a copy every 45 degrees).
	**Copy as it is**	Click a wall with its windows and doors, then the walls to copy them onto - or a room, then the rooms to copy everything in it into.
	**Mirror**	Flips what is selected left to right or front to back.
	**Paint bucket** `B`	Pick a colour or material, then click anything - a wall face, a floor, a roof, an object - to paint it. Alt + click picks up the paint from what you click.
	**Group** `Ctrl+G`	Ctrl+click several things, then Group (Ctrl+G): they move, turn, copy and delete as one. Ctrl+Shift+G takes a group apart.
	**Delete** `Del`	Deletes what is selected (Delete key). Ctrl+Z brings it back.
	**Eraser**	Click walls, doors, windows, objects or guide lines one after another to remove them.
	**Fit a door or window**	Draw a box over the gap between two walls (or across a wall) and pick a door or window: it is made to fill exactly that space.
	**All walls**	Select the wall you have set up, then All walls: put its height, its thickness, both, its material, colour and build-up, or everything on every wall at once - this floor or the whole house. With no wall selected (or any time) type one height or thickness for all walls. Doors and windows that would no longer fit a lower wall are made to fit. Ctrl+Z undoes it.
	**Open/ close**	Click windows and doors one by one to open or close them (Shift + click: half open), or open, half open or close all of them - or only the windows, or only the doors. Doors swing into the house, casement windows swing out on their side, awning and top-hung windows swing out from the top, sliding ones slide, a garage door rolls up. Fixed glass stays shut.
	**Extend or trim**	Click near the end of a wall: it runs on to the next wall it meets, or is cut back to it.

	**Curve**	Select a wall or a box, then Curve: bend it, or change a curved wall's radius, flip it, straighten it, put curved windows or a curved glass wall across it, or a door.
	**Join walls**	Click two walls: their ends are made to meet in a clean corner.
	**Chamfer**	Cuts a corner between two walls off with a short wall across it.
	**Round corner**	Makes the corner between two walls a curve.
	**Parallel wall**	A copy of the selected wall at the distance you give, side by side with it.
	**Match**	Click the wall or object to copy from, then the ones to change: they get its height, thickness, material and colour.
	**Pick paint**	Click anything to take its paint; then the Paint bucket paints with it.
	**Align**	Ctrl+click two or more things, then pick how: sides, fronts or backs in line, middles in a row, spaced evenly, tops or bottoms level, all on the floor, stacked, or all turned the same way.
	**Stick together**	Click the things that must always move together and press Enter (or click a box: everything resting on it is stuck to it). Moving, turning, copying or deleting one takes all. Unstick takes them apart.
	**Check and fix**	Looks over the whole plan for walls that do not meet, walls drawn twice, doors and windows that overlap or do not fit - lists them and fixes them for you.

Tool reference: Free-form

Curved and flowing shapes: domes, waves, lofted surfaces, and dots to shape anything.

	Tool	What it does and how to use it
	**Shell**	Click two corners on the ground and type a height: a dome, a vault, a wave or a sheet. Then shape it with its dots.
	**Loft**	Draw curves on the plan, each at its own height; a smooth surface runs through them - curved walls, flowing roofs, whole buildings.
	**Edit dots**	Select a free-form shape, then drag its dots up or down (Shift + drag moves a dot sideways).
	**Dots on anything**	Click any shape (a box, a roof, furniture), then press where you want a dot and drag: the surface round it follows. R sets how far round the dot the shape moves.
	**Shape it**	Bend, twist, taper or smooth a free-form shape; give it thickness, panels or a glass band.

	**Shape bend**	Click the object (A), then draw the line it should follow (B): it bends along it - a bench, a railing, stairs, a window.
	**True bend**	Click the object (A), then where it goes (B) - an opening, a face or a curved surface: it is fitted to its edges or bent onto it.
	**Window or door as object**	Click a window or door in a wall to turn it into an object you can bend, fit or paint; click an object standing in an opening to make it the wall's own window or door again.
	**Bend dots**	Drag the dots along a bent object's line to bend it differently.
	**Example**	Builds an example: a flowing building with a curved glass front, to show what these tools can do.

Tool reference: Solid tools

Join and cut solid shapes. Select two shapes (Ctrl+click) first.

	Tool	What it does and how to use it
	**Union**	Select two shapes: they are joined into one solid.
	**Subtract**	Select two shapes: the second is cut out of the first (a hole, a notch, an arch).
	**Intersect**	Select two shapes: only the part where they overlap is kept.
	**Trim**	Select two shapes: the first is cut out of the second, and both are kept.

Tool reference: Build

The parts of a building and its surroundings: stairs, roofs, railings, ceilings, bridges, roads, ramps, decks, retaining walls, furniture and models.

	Tool	What it does and how to use it
	**Stairs**	Click where the stairs start, then where they arrive upstairs: the number of steps and their size are worked out, and the hole in the floor above is cut.
	**Roof designer**	Opens the roof plan: the roof from above with every line in its colour and name (ridge, valley, hip, eaves, verge ...) and its length, with the tools to change the roof style, draw roof lines, set pitch, overhang and covering, and add skylights, chimneys and vents. Done or Esc goes back.
	**Bridge**	Click where the bridge starts and where it ends, pick beam, arch or cable-stayed, over a valley or a flyover, the lanes and the height. Deck, road, lines, barriers, piers and abutments are made as one group.

	**Path and Road**	Click points along the ground where it goes, then Enter. Pick a road (one or two lanes, with kerbs and white lines), a driveway, or a paved, stone or gravel path, and type how wide. Corners are rounded; on a slope it follows the ground.
	**Slope / Ramp**	Draw a ramp: click the bottom, then the top, type how high and how wide - railings both sides, the gradient checked for a wheelchair (1 in 12). Or select something first and Tilt it: click the end that stays, then the end that rises, and type how much.
	**Round balcony**	Click a round wall, pick a deck at the floor or a balcony at the next floor, how far out, and right round, half or a quarter. Railing and posts are added.
	**Retaining wall**	Click points along the slope, then Enter. Pick concrete, stone, brick or blocks, the height, and whether its top follows the ground or steps down level.
	**Round house**	Click the middle, then the radius, and pick a look: a whole round wall with a door and two windows under its own round thatch, tiled or metal roof.
	**Curved stairs**	Click the centre they turn round, then where they start, move round and click where they arrive - any turn, even a full spiral.
	**Stairs along a path**	Click where the stairs start, where they bend and where they arrive: S-curved or flared stairs along your path.
	**Curved edge and steps**	On the top of a terrace or slab, click where the curve starts, where it ends, bend it and click: the corner is cut off along the curve.
	**Build tools**	All the Build tools in one list.
	**Railing**	Pick a type (glass, bars, rails, cables, slats, wrought iron, picket, panel), then click point after point; Enter finishes, the first point again closes it. Height, posts and colours are in Settings.
	**Roof of your choice**	Pick a roof type (hip, gable, shed, flat, gambrel, mansard, pyramid) and click the corners of what it covers - a garage, patio or carport.
	**Roof lines**	Click the eaves all round, then draw the ridge (a peak, a line or a ridge that bends) and type its height: the roof is made exactly as drawn.
	**Roof structure and pergola**	Timber or steel trusses, rafters and purlins under a roof, or a pergola or carport frame over an outline you click.
	**Ceilings**	Pick flat, bulkhead, tray, LED cove, cornice, coffered or sloped - in every room, the selected room, or over an outline you draw.
	**3D text**	Words standing up or lying flat, bold or regular, any height and thickness. Make them glow in Settings for a sign.
	**Furnish**	Fills every named room with furniture that fits it (beds in bedrooms, a dining set ...) and a ceiling light. Things you placed yourself are never moved; it can be taken out again.

	Add a part	Adds a part: a wall, door or window, from a list.
	Library	Furniture, kitchens, bathrooms, lights, electrical and plumbing points, cars, people, trees and plants: pick one and click where it goes.
	Online library	Free furniture, doors, windows, plants and materials from the online library, suggested room by room for your plan.
	Import a 3D model	Bring in a 3D model you have (.obj, .glb, .gltf, .dae, .stl); it becomes an object you can place, move and paint.
	Packs	Add a pack of models, materials and ready houses (a .zip), or make a pack of your own to share.

Tool reference: Modern

A modern look in a few clicks, and the tools for its details.

	Tool	What it does and how to use it
	Nestify Studio	Opens the selected thing (the roof, a shape, furniture, a wall) in Nestify Studio to design it freely on its own, with everything else locked. Done brings it back.
	Make it modern	Turns the house modern in one go: flat dark roof with a deep overhang, white walls, slim black frames, a stone entrance wall and a pivot door under a canopy. Ctrl+Z undoes it.
	Feature panel	Click two corners on a wall's face: that area gets stone, timber, brick, concrete or dark render.
	Canopy	Click two corners on the plan: a concrete slab, curved or arched metal, or glass canopy over a door, terrace or carport.
	Screen	Click where it starts and ends: timber slats, aluminium fins or louvres.
	Corner glass	Click a corner of the house: both walls get glass to the corner, with no post.
	Double volume	Click a room: the floor above it is opened up, so the room is two storeys high.
	Window & door designer	Click a window or door to redesign it (panes, bars, frames, glass, how it opens), or a wall to design a new one for it.
	Modern lights	Adds downlights in every room, warm LED strips under eaves and canopies, and uplights on feature walls.

Tool reference: Facade

Details on the outside walls: parapets, cornices, plinths, pilasters, surrounds, porch steps, columns

and grooves.

	Tool	What it does and how to use it
	**Parapet and cornice**	A low wall round the top that hides a flat roof, plain or with a stepped cornice - for the whole house, or only the selected walls at their own height.
	**Plinth**	A raised band along the bottom of the outside walls; it stops at the doors.
	**Pilasters**	Raised strips at the outside corners, or where you click, plain or with grooves.
	**Window surround**	A raised frame round windows and doors: thin, wide, a deep box or a sill - all of them, or click the ones you want.
	**Porch steps**	Click two corners of the platform, type its height, and pick steps on the front, front and sides, or all round.
	**Columns**	Pick a column - classical with base and capital, twin, fluted, round or square - and click where it stands.
	**Groove**	Click where the line starts and ends on a wall: a groove is cut into the plaster.

Tool reference: Measure and view

Measuring, zooming, hiding and walking through.

	Tool	What it does and how to use it
	**Tape measure** `E`	Click two points: the distance shows, and can be kept on the plan as a measurement.
	**Sizes on / off**	Shows or hides the sizes written in the 3D view.
	**Zoom in**	Zooms in (the mouse wheel does too).
	**Zoom out**	Zooms out.
	**Zoom window**	Click two corners of the area to zoom to it.
	**Frame** `F`	Fits the whole building in the view.
	**Top view** `T`	Switches between looking straight down and the 3D view.
	**Measure all**	Writes dimension lines along every wall, door and window of the plan for you.

	Show only this	Shows only what is selected; Show everything brings the rest back.
	Show all	Brings back everything you hid.
	Hide	Hides what is selected (it is still there, and still counted).
	Walk through the house `W`	Walk through the house at eye height, starting at the front door: W/S or the up/down arrows walk, A/D step sideways, Q/E or left/right turn, drag to look, Shift walks faster, click the floor to walk there. Doors swing open; a mini-map and a Rooms list help you get round. Esc stops.
	Reference picture	Puts a photo of the house you want on a card beside the model, to copy from: drag it, size it, zoom it, close it.
	Lock all	Locks everything so nothing moves by mistake while you look round; press again to unlock.
	Remove guides	Removes every guide line on this floor.

Tool reference: Scene

The plot, the ground, the sun and sky, pictures, videos and the cost.

	Tool	What it does and how to use it
	Site	Makes the plot slope: how steep, which way it falls, and the level platform the house stands on.
	Dig	Dig a hole by its outline or as a rectangle, or the foundation trenches under the walls (with a concrete footing). The sides show topsoil, sand, clay and rock. Fill a hole in by clicking it.
	Hills and dips	Click the ground to put a dot: it rises into a round hill (press Dip first and it sinks). Click a dot to change how high and wide it is, move it or delete it. The house keeps its level platform.
	Sun and shadows	Sets the time, month, place and north, so shadows fall where they really will.
	Sky	Clear, a few clouds, cloudy, overcast, sunset, dusk - or your own sky picture.
	Tiles and stone	Tiles and stone for floors and walls: marble, mosaic, flagstone, slate and more.
	Describe the picture	Type what the picture should look like (evening, warm lights, cloudy, modern ...): the sun, sky, lights and finishes are set, then a photo-real picture is offered.

	Walk video	Click the start (the front door), then the rooms in order, then Enter: an MP4 video walks through the doors and looks round each room.
	AR page for phones	Makes a page for phones: open it and see the house on a table, or full size on the real plot.
	Quantities and cost	The full bill of quantities and cost (bricks, cement, sand, concrete, roof, doors, windows, finishes, site works ...) with prices you can change. The running total is in the status bar.
	Draw the plan in 2D	Opens the 2D plan to draw walls, doors, windows, rooms, railings and roofs from above, in four looks.

Tool reference: More tools

Tools that are not on the rail: the hand next to the Find a tool box, the Section tab at the top, Exact numbers (find it with Ctrl+F) and Rescue in the project menu.

	Tool	What it does and how to use it
	Hand `H`	Drag to slide the view sideways (right-drag and the arrow keys do too).
	Section	Cuts the house open with a plane you can move, to look inside or print a section.
	Exact numbers	Type the exact position, size and angle of what is selected.
	Rescue a project	In the project menu: opens a project that will not open normally anyway, and shows what is wrong in it.

Keyboard shortcuts

The keys work in the 3D view when you are not typing in a box.

Key	What it does
`Space`	Select tool
`Esc`	Stop the tool / back to Select
`L`	Wall
`N`	Line
`R`	Rectangle
`C`	Circle
`U`	Shape push/pull
`D`	Opening
`M`	Move

`Q`	Rotate
`P`	Push/Pull
`S`	Split
`B`	Paint bucket
`E`	Tape measure
`F`	Frame the building
`T`	Top view / 3D view
`W`	Walk through the house
`H`	Hand
`[ ]`	Fold the side panels
`Del`	Delete the selection
`Enter`	Finish (outlines, paths)
`Ctrl+Z`	Undo
`Ctrl+Y`	Redo
`Ctrl+S`	Save
`Ctrl+O`	Open
`Ctrl+F`	Find a tool
`Ctrl+G`	Group
`Ctrl+Shift+G`	Take a group apart
`Ctrl + click`	Add to the selection
`Alt + click`	Pick up paint (with the bucket)

While drawing or moving, just type a number (or two, like 4,3) and press Enter for an exact length, size or distance.

Nestify Studio (design anything freely)

Select something and press **Studio** (the sparkle button at the top, **Open in Studio** on the right, or **Studio** on the roof's card):

- **The roof** comes in as a **solid** you can shape freely (cut, pull up, add dormers, make parts glass), or as a **roof** you can still restyle (hip, gable, Dutch gable, butterfly ...). The house roof hands over to it; Ctrl+Z puts the house roof back.
- **A shape, furniture, a group, or several things Ctrl+clicked** come in as they are.
- **A wall** comes in as a shape copy with its doors and windows cut out; the wall itself is hidden while the copy takes its place (Ctrl+Z brings the wall back).

- **Nothing selected:** you start from a 1 m block in front of you.

The rest of the house turns to faint lines and the **Studio card** on the left holds every tool:

Group	Tools
Draw	Rectangle, Circle, Polygon, Line, Arc, Sweep, 3D text, New block
Shape	Shape pull (any outline on any face, pushed in or pulled out), Push/Pull, Move, Rotate, Scale, Offset, Mirror, Copies, Free shape (makes furniture or a roof one shape every tool can cut)
Combine (Ctrl+click two)	Join, Cut out, Overlap, Trim
Roof	Dormer (click a slope: front wall, window and its own little roof), Covering, Roof colour
Look	Clear, tinted or frosted glass, Glowing, Metal, Standing seam, Timber, Stone, Brick, Concrete, Plain, Colour, Paint, Your picture
Keep	Save to My items, Copy, Delete

Everything you make joins the design. **Done** (or Esc) goes back to the whole house, and the design stays where it is in it.

Window & door designer

Press **Designer** (MODERN group on the left, or **Window & door designer** on the Studio card), then click a window or door in the house to change it, or click anywhere on a wall to put a new one there. The designer card (right of the view) changes it while you watch:

Part	Choices
Type	Window or door
Size	Width, height and (windows) sill height: - / + in 100 mm steps, or click the number to type it
How it opens	Fixed, casement (hinged), double, sliding, folding / stacking, pivot
Panes	Panes across and up
Glazing bars	None, grid, colonial, lines across, lines up
Top light	None, small, medium, tall (a fixed pane over the leaves)
Frame	Thin, medium or chunky; black, charcoal, bronze, silver, white or timber
Panel	Glass, half glass (solid lower part), solid, slatted; and the panel colour
Glass	Clear, grey, bronze, blue, green, frosted, or no glass

Save keeps the design under **My designs** (on the same card) so you can put it on any wall with one click; **All like it** copies it to every window or door of the same size on this floor. A designed window or door is a real opening in the wall: the plans, the cost and photo-real pictures use it. Ctrl+Z undoes the

whole edit.

Roof styles and coverings

Click the roof and press its style button: **hip, gable, Dutch gable** (a hip with a small gable at the top of each end), **butterfly** (a V with the valley in the middle), **pyramid, shed** / **skillion, flat, gambrel, mansard**, **barrel** (arched) or **curved lean-to** (bullnose). Every style follows the plan: L, T and U shaped houses get valleys where the wings meet. **Material** on the right picks the covering: clay or concrete tiles, IBR, **standing seam, corrugated, tile-effect metal**, slate or **shingles**, and **Roof colour** the usual colours (charcoal, black, slate grey, galvanised, barn red, terracotta, forest green, ocean blue, brown, copper, teal, cream) or any colour. A roof you draw yourself (Roof on the left) takes the same styles over the outline you click. For anything more (dormers, cut-outs, glass panels), take the roof into the Studio.

Roof measurements

Click the roof, then Measurements (on the right; in the 2D plan's roof card: All roof measurements; in the Studio: ROOF > Measurements). You get a plan of the roof with every line in its colour and length (ridges, hips, valleys, pitch changes, eaves, verges), the real sloped area and the area it covers, the pitch and the longest rafter. Below is what it comes to: tiles or sheets (with their lengths), ridge and hip caps, valley flashing, gutters and fascia, downpipes, barge boards and battens or purlins. Count as switches between tiles, slate, shingles, IBR, corrugated, standing seam and tile-effect sheets to compare. Copy puts it all on the clipboard for a quote or a supplier. Roofs you draw with the Roof tool and roofs reshaped in the Studio are measured too, and Quantities and cost counts them.

Curved roofs and canopies

Two curved roof styles: Barrel (an arch from eave to eave) and Curved lean-to (a bullnose: it rises from the low eave and levels off at the top). Pick them like any roof style; the pitch sets how steeply the curve meets the eaves. Canopies (MODERN > Canopy): click two corners, pick the height, then the kind: flat concrete slab, curved metal (bullnose), arched metal (barrel), curved glass, or flat glass on steel arms. Curved canopies put their high side against the wall. A carport canopy stands on steel posts where it is not against a wall. Canopies are roof pieces: Settings changes their style, curve and covering, and Measurements gives their sheets.

Round and bevel edges

In the Studio (SHAPE): click a piece, then Round edges or Bevel edges. Choose which edges: all of them, the top edges and corners (the bottom stays sharp so it still sits flat, good for countertops, parapets and caps), or the upright corners only (a soft wall end or column). Then type the size in mm. Inside corners stay sharp and the piece never grows; a size too big for the piece is refused. Ctrl+Z puts the edges back.

Roof lines (draw your own roof)

 Roof lines (BUILD on the left, or ROOF in the Studio) makes a roof exactly as you draw it, for roofs no style button gives: a ridge off the middle, a ridge that bends, a peak in a corner. Click the roof's outside corners (the eaves, overhang included) all round, then the first corner again or Enter. Then click the ridge: one point for a peak, two for a ridge line, more for a ridge that bends, and Enter. Type how high the ridge is above the eaves (it suggests the height for a 30° pitch). Every face runs straight from the eaves to the ridge; where the ridge bends you get a valley inside the bend and a hip outside it. When it covers the house it replaces the automatic roof. It is measured and costed like any roof, and you can open it in the Studio for dormers or glass. Draw an L or U shaped house as one roof per wing.

Exact numbers

Exact (on the right when a piece is selected, or SHAPE > Exact numbers in the Studio) opens every number of the piece: where its middle is (X and Y on the plan), how high it is raised, its width, depth and height, how far it is turned, tipped forward and rolled sideways, and Move by X / Y / up to shift it an exact distance. Type in your unit or with one (2400mm, 8ft, -1.5) and Apply; one Ctrl+Z puts it all back. After a turn with the Rotate tool's circle, type an angle and press Enter: the turn becomes exactly that. After a move, a Push/Pull or Copies, typing a number and Enter works the same way.

Curved stairs

 Curved stairs (BUILD on the left, or STAIRS in the Studio) are set out the way builders do it, round a centre: click the centre the stairs curve round, then where they start - how far from the centre is the middle of the stairs (the walk line) - then move the mouse round the centre. The steps show as you go, each pointing at the centre, with how many there are, how high and how deep on the walk line (it warns when they get too steep); go as far round as you like, past a full turn for a spiral. Click where they arrive, or type the angle (like 180) + Enter. They are made as a smooth concrete flight (one sloping slab under the steps, no stepped underside) with glass on both sides and a handrail. Stairs along a path (next to it): click where they start, points where they should bend, and where they arrive - for S curves and sweeping flared entrance stairs. Settings on the right changes everything afterwards: the width, how much wider the bottom steps flare out and over how many steps, the height (to the floor above or any height), the number of steps; how they are built (a smooth concrete flight, solid to the ground, side beams with closed or open risers, floating treads), timber treads, each side a solid wall, glass, bars or nothing, and one or two landings. They are walked on, cut their own hole in the floor above, show on the plans with their treads and UP arrow, and the building rules check their steps.

Free-form shapes (FREEFORM)

For flowing buildings, swooping roofs and curved glass walls. Shell: click two corners on the ground and type a height - a dome, a vault (an arch), a wave or a flat sheet. Loft: draw a curve on the plan (click points, Enter), type its height, then the next curve and its height ... Enter on an empty curve

makes a smooth surface through them all: two curves make a curved wall, more make a flowing roof or a whole building. Edit dots: the surface's dots show - drag one up or down, Shift + drag moves it sideways; press anywhere else on the surface for a new dot right there (as many as you like); the surface always passes smoothly through its dots. Shape it: bend it round, twist it as it rises, taper it, smooth it, give it a thickness, panel lines over it (like cladding panels, they follow the curve) and a glass band (a part of it glass, with its frames following the shape - the glass front of a building). It all stays editable. Example puts a flowing building with a curved glass front in the view, to see and take apart.

Shape bend and True bend

Both work with A and B: A is the object (from the library, a window, a door, furniture, a shape you made - click it first, or click it after starting the tool), B is where it goes.  Shape bend (FREEFORM): then click points along the line B it should follow on the plan and press Enter: the object bends along it from end to end - a bench round a curved terrace, a window or railing round a curved front, a planter, even a flight of stairs. It keeps its own parts and colours (glass stays glass). Bend dots then shows the line's dots: drag one to bend it differently, press on the line for a new dot there. True bend: click B, and Nestify finds its edges and fits the object to them. A window or door opening in a wall: the object takes the opening's width and height and sits in it, lined up with the wall (the opening's own window makes way). A flat face (the side of a wall or a shape, a top): the object takes the face's width and height and sits against it. A curved surface (a free-form shell, a rounded shape): the object keeps its size and is bent to lie on the surface, following its curve.

Curved walls

 Curved wall (SELECT AND DRAW): to bend a wall you have, click its two ends, move the mouse to bend it and click (or select it and use Curve): that same wall curves, keeping its material, windows and doors. For a new one, click where it starts, then where it ends, then move the mouse to bend it and click; started on the end of a wall, it takes the material, height and thickness of that wall. Or click any straight wall and  Curve (CHANGE): Bend it (how far the middle sits off the straight line; 0 makes it straight again), Radius, Flip, Make it straight. A curved wall is one smooth solid wall (and a window along it one bent pane of glass): clicking any part of it selects and lights up all of it, it moves, hides, locks and deletes as one, the plan, the list and the costs follow the curve, and Length shows the length along the curve. Windows and doors stay with it when it bends: a window follows the curve (curved glass), a door bends with it too (a curved leaf, its handle near the end it opens from). Curve also adds a curved window across it, a floor-to-ceiling curved glass wall, or a door. Doors and windows (2 points or 4 corners) on a curved wall follow the curve too: a window becomes curved glass from the first point to the last, and so does every door: a glass slider or stacker as curved glass, a hinged door as a curved leaf. With the Curved wall tool, clicking twice on the same wall (anywhere on it, even its top) bends that wall. Every door and window choice shows a picture of it (the Door and Window buttons, Door/window and the Doors and windows card). True bend on a curved wall bends the object round it: into a curved opening, or standing against the side you clicked. Boxes and shapes bend the same way: with the Curved wall tool click on a box (a Rectangle pulled up, a box used as a house, any shape or object) where the curve starts - that is the first dot - then click where it ends (the second dot; the dots sit on the middle of the box's length, and a dot shows where each click will go),

move the mouse to bend it to either side and click. Only the part between your dots curves; the rest stays straight, and its width, height and the doors and windows cut into it are kept. Do it again elsewhere on the same box for an S-shape or a wavy wall. Curve (CHANGE) on a selected box bends it by a typed amount, flips the curve or makes it straight again.

Windows and doors as one family

A window or door made with the Door and Window buttons is part of its wall; one from the library is an object.  Window or door as object (FREEFORM) turns one into the other: click a window or door in a wall and it becomes an object standing in the same hole - the same look, but now it can be bent (Shape bend), fitted (True bend), painted, moved or edited with Dots. Click an object standing in an opening (or select it and press the button) and it becomes the wall's own window or door again. Shape bend and True bend do this by themselves: click a wall's window or door as A and it is used like any library window.

Push and pull one side of an object

Push/Pull now finds the outline of the side under the mouse, not only the whole box. Hover a door frame and only the jamb you point at lights up: drag it (or type a distance) and just that side moves out or in - one leg of a frame thicker, one step deeper, the inside of a window reveal. On a plain face the whole side moves as before.

Smoother dragging

While you drag dots (Dots, Bend dots) Nestify works out only the newest mouse position once per frame, skips the other storeys and the roof until you let go, and redraws the shape in one pass, so the shape keeps up with the mouse on slower PCs and big houses.

Facade details (FACADE)

The tools on the FACADE group of the rail add the details that make a house look finished, like the flat-roof houses with stepped cornices, banded corners and framed windows. Each one wraps the corners of the house by itself, and each piece is an ordinary object you can paint, move or delete; running a tool again replaces what it made. Parapet and cornice: pick a style from the pictures (plain, with a cap, a 2-step or 3-step cornice, or a cornice with no parapet), type the parapet height and pick its colour; it goes round all the outside walls and the roof turns flat behind it. Select some walls first to give that part of the house its own height (a taller front block). Plinth: a raised band along the bottom of the outside walls, stopping at the doors. Pilasters: raised strips at every outside corner or where you click, plain or with grooves across them. Window surround: a thin trim, a wide frame, a deep box frame (the bold coloured frames) or just a sill, round every window, every window and door, or the ones you click. Porch steps: click two corners of the platform, type its height and pick the sides that get steps (front, front and sides, all round). Columns: classical (base and capital), twin, fluted, plain round or square, then click where it stands. Groove: click where a line starts and ends on a wall and it is cut into the plaster (Shift keeps it straight up or across). The Awning window (Window button, Doors

and windows card) shows its panes hinged at the top and opened out.

Round houses and thatch

 Round house (BUILD): click the middle, then how far out the wall goes (or type the radius) and pick a look - a modern rondavel (plaster walls, thatch), a traditional one (mud walls, thatch), or plaster walls under a round tiled or metal roof. You get one whole round wall (a real smooth circle) with a door at the front and two windows, and its own round roof with a cap on the point; it can stand next to a square house without touching that house's roof. Click the roof to change its pitch, overhang, thickness or covering. Any house roof can be Round (a cone that follows the walls all the way round) or a Dome too, and Thatch (grass) is a covering for every roof (Quantities count it in grass bundles). Mud (daga) is a wall material, and circles and cylinders you draw are smooth now (no flat sides).

Bridges

 Bridge (BUILD): click where the bridge starts and where it ends, then pick the kind - a beam bridge (a deck on piers), an arch bridge (stone arches) or a cable-stayed bridge (a pylon in the middle holding the deck with cables) - where it goes (over a valley or river: the ground is dug out under it and a river runs through; or a flyover with a ramp up at each end), how many lanes (1, 2 or 4) and how deep the valley or how high the deck is. You get the deck with its road, white lines, pavements, concrete barriers and rails, the piers with their cap beams and the abutments at the ends - one group that moves, turns and copies as one.

Drawing on any face and pulling out new pieces

Rectangle, Circle, Polygon and Line work on any face, not only on the ground and tops: start one on the side of a box, a shape or a wall and it is drawn on that face exactly where you click (the tool switches to Shape push/pull with that shape), then push it in (a niche, a recess, a hole) or pull it out. Pulled out - with the mouse, with Push/Pull (P) or by typing a distance - the outline becomes a new piece of its own, standing against the face it was drawn on: it is selected straight away and Push/Pull, Move, Rotate, Paint, Split and every other tool work on it alone, so a wardrobe, a band, a ledge or a block on top is built piece by piece;  Union (SOLID TOOLS) joins pieces into one when you want them as one. Drawn on the top of a wall, it is a new block standing on the wall. Push/Pull now shows every push and pull while you drag (before, a push into a shape could look as if nothing happened until the view was rebuilt). The mouse shows the tool you are using. With a drawing tool (Line, Wall, Rectangle, Circle, Shape push/pull, Curved wall, roads ...) the pointer is a pen: its nib is exactly where the line starts - on the corner or edge it snaps to - and the line you draw runs from the nib. With the other tools their icon (the eraser, the paint bucket, Push/Pull ...) sits beside the pointer, and the pointer is a hand while you slide the view. Up 1 cm / Down 1 cm on an object's card lift or lower it in small steps (Raised above the floor takes any typed height). For the dark banded corners and the stepped band round the top in a modern flat-roof house, use Pilasters (raised corner strips with grooves across them) and Parapet and cornice in FACADE; Shape bend makes any object follow a line you draw, round corners too; Stick together keeps things together.

Apply to all walls

 All walls (CHANGE, the button with the select-all icon): set up one wall the way you want it (its height, thickness, material, colour), keep it selected and press All walls. Pick what goes on every other wall: its height, its thickness, height and thickness, its material, colour and build-up, or everything. With more than one floor it asks: every wall on this floor, or every wall in the whole house. Or type one height or one thickness for all walls (no wall needs to be selected). Doors and windows that would no longer fit a lower wall are made to fit. Locked and hidden walls are left as they are. Ctrl+Z undoes it.

Opening windows and doors

Open / close (CHANGE): every window and door that opens can stand shut, half open or wide open, and it shows in 3D, in pictures and photos and on the walk. Click windows and doors one by one (each click opens or shuts it; Shift + click opens it half way), or open, half open or close all of them at once - or only the windows, or only the doors. Hinged doors swing into the house; casement windows swing out on their side; awning and top-hung windows swing out from the top; sliding doors and windows slide over their neighbour; a garage door rolls up; a pivot door turns on its pin. Picture windows and frameless glass are fixed and stay shut. In the Window & door designer, Open it (Closed, Half open, Wide open) does the same for the one you are designing, and Top-hung is a way of opening of its own. Awning windows come in two kinds (Window button, Doors and windows card, each with its picture): Awning (a grid of panes all hinged at the top) and Awning + side light (a fixed pane down one side, the rest top-hung), both with charcoal frames; they stand a little open until you close them. When you walk through the house every door stands open.

Roads and paths

 Path and Road (BUILD): click points along the ground where it goes, then Enter (or click the last point again). Pick a road (one lane, or two lanes with white edge lines and a dashed centre line, both with kerbs), a concrete driveway, a brick-paved path, a stone path or a gravel path - each shows as a picture - and type how wide. The corners are rounded smoothly and on a sloping plot it follows the ground, point by point. It is one group (road, kerbs and lines move together). The estimate counts the asphalt or paving in m², the kerbs and painted lines in metres.

Slopes and ramps

 Slope / Ramp (BUILD): Draw a ramp - click the bottom, then the top, type how high the top is (it tells you the most a wheelchair can manage: 1 in 12) and how wide. You get the ramp with a level landing at the top and steel railings on both sides, and it says the gradient it has. Or select an object first (a road, a path, a driveway, a box, a bridge approach) and pick Tilt: click the end that stays, then the end that goes up, and type how much higher - it rises evenly from one end to the other (minus: it goes down). A road tilts with its kerbs and lines.

Round balconies

 Round balcony (BUILD): click a round wall (from Curved wall or Round house), then pick a deck at the floor (a little above the ground) or a balcony up at the next floor, how far out it reaches, and right round, the front half or a quarter at the front. You get the curved floor, a curved railing with posts and rails and, when it is up off the ground, posts down to the ground - one group. The estimate counts the deck in m² and the railing in metres.

Retaining walls

 Retaining wall (BUILD): click points along the slope where the wall goes, then Enter. Pick concrete, stone, face brick or concrete blocks, type how high it stands above the ground, then its top: following the ground (always as high above it, with a coping on top) or level, stepping down the slope every 2.4 m in 150 mm steps. It goes 400 mm into the ground. The estimate counts the face in m².

Plans and costs of the new tools

The Floor Plan, the PDF plans, the DXF file and the council pack now show curved walls as real curves (true arcs in the DXF, the radius written beside them) with their doors and windows, round roofs by their eave and point, the holes and foundation trenches you dug (dashed, with how deep), and roads, paths, ramps, retaining walls, decks and bridges by their real outline. The council pack gets a Foundation plan page when you have dug foundations, and the site plan counts every roof (a round house too) in the coverage. The estimate adds: digging (m³ per hole and for the trenches), the footing concrete you dug (instead of the rule-of-thumb footing), cornices, plinths, pilasters and window surrounds (metres), parapets (m²), porch concrete and tiles, columns, the bridge (concrete, stone, steel, asphalt), roads, paths, kerbs, road lines, ramps, balconies, railings and retaining walls - each with a price you can change.

Digging the ground

 Dig (SCENE): a hole by its outline (click its corners, then Enter), a rectangular hole (two corners), or the foundations from the house plan (a trench under every ground-floor wall, each as wide as its wall plus a margin each side, as deep as you say, with a 300 mm concrete strip footing in the bottom if you like; dig them again after changing the plan and they follow it). While you draw a hole, the house plan (the walls) shows on the ground so you can lay it out against it. Type how deep and it is dug: the sides of the hole show what the ground is made of, layer by layer, as deep as it goes - topsoil (0 - 0.3 m, dark earth), sand (0.3 - 1.2 m), clay (1.2 - 2.5 m) and rock below that - and the bottom shows the layer it reaches. Ground layers changes how deep each layer goes (for your site's soil report). Fill a hole in (click it) takes it away again, footing and all. Holes work on a flat plot and on a sloping one. To see into trenches under the house, hide the floor slab (its eye in Project Structure).

Roof designer

Roof designer (BUILD): the roof from above - a roof plan - with every line in its colour and name, and its length: ridge (red, dashed), valley (blue, dashed), hip (green), roof edge / eaves (dark blue), verge (purple), pitch change (cyan) and high edge (yellow); thin rafter lines run down each slope, and the front door shows as a black triangle. The legend on its card says which line is which and how long they are in total. Its tools each show the colour of what they make: the roof style of the whole house, a roof drawn by its edge, the ridge drawn yourself (Roof lines), pitch, overhang, covering, skylights, chimneys and roof vents (click on the roof), and the measurements and materials. Names on the lines and the rafter lines can be switched off. Done (or Esc) goes back to the 3D view as it was. The Roof measurements window uses the same colours.

Align and stick together

Align (CHANGE): Ctrl+click two or more things, then Align: their left sides, right sides, fronts or backs in line; their middles in one row (left to right, front to back, or one straight row at any angle); spaced evenly, or with the same gap between them; their bottoms, tops or middles level; all standing on the floor; stacked one on top of the other; or all turned the same way as the selected one. A group (or things stuck together) lines up as one. Stick together (CHANGE): click the things that must always move together (boxes, furniture, walls) and press Enter - or click a box and everything resting on it (and on those) is stuck to it - so a box on top of another can never be moved off it by mistake: moving, turning, copying or deleting any of them takes all of them. Unstick takes them apart. Floors move together (on the same card): moving a wall, or dragging a corner, moves the walls right above and below it on the other floors too, so the storeys of a double-storey house stay one on top of the other.

Dots on anything

Dots (FREEFORM), or Edit dots on any shape: click a shape (a box from Rectangle, a roof, furniture ...), then Dots. A net of dots with lines between them covers the side facing you - a square grid on a box, small even triangles on rounder shapes. Drag any dot out of the face or into it; Shift + drag moves it any way across the screen. The shape follows; R sets the reach - how far round a dot the shape moves with it (0: just the dot). Turn the view and the dots on the other side show. Esc stops. A roof shaped this way becomes a free shape (still measured and costed as a roof). Shape it on a solid shape also bends, twists, tapers or smooths it (Ctrl+Z undoes it).

Curved edge

Curved edge (BUILD): click on the top of a terrace, platform, slab or any shape where the curve starts (near an edge), then where it ends (on another edge), move the mouse to bend the curve (it shows as you go) and click. The corner beyond the curve is cut off straight down, so the side follows the curve. Steps along it are drawn afterwards like any stairs (Curved stairs, round the same centre).

Find a tool (Ctrl+F)

The Find a tool box sits at the bottom left of the 3D view, next to the hand. Type what you want to do - stairs, cut, copy, roof, measure, rescue - and the matching tools show above it as you type; click one (or Enter for the first) and it starts. Ctrl+F puts the cursor in it. Other words work too: "spiral" finds the curved stairs, "broken" finds Rescue.

Tips and the view buttons

The Tips card (bottom right of the 3D view) shows what the tool you picked does and what to do next; a long one scrolls. Bottom left of the 3D view: the hand and the Find a tool box. Measuring, zooming, the top view and framing are in MEASURE AND VIEW on the left. Section is the tab at the top.

Moving and resizing cards

Every card over the 3D view - Doors and windows, What's next, Camera, Sun, Site, Dots, the Online library, the tips card and the selection card - can be moved and resized, so it never sits where you are working. Drag the short bar at its top (or any empty part of the card) to move it anywhere on the view. Drag its bottom right corner to make it narrower or wider (and taller, where it has a list to show); names on its buttons shorten to fit. Double-click the bar to put it back where it was. Each card opens where you left it next time. H or C still tucks all cards away at the top right.

Tucking cards away (H, C)

The cards on the 3D view (camera angle, sun, push/pull, the designer ...) step aside when you press H (the hand) or C: they become small chips at the top right of the view. Click a chip and its card comes back exactly where it was, with its choices as they were.

Editing one thing safely

While you edit one thing on its own (Edit group, Open in Studio), everything else in the house - every wall, the roof and the other objects, on every floor - is locked, so nothing can be moved, cut or deleted by mistake; the locks show in Project Structure. Done (or Esc) unlocks them all again; anything you had locked yourself stays locked.

Locking things

Each wall, the roof and every object has a lock beside its eye in Project Structure; the lock on the house's row locks everything at once (Lock all on the left does the same). A locked thing can still be clicked, but dragging on it turns the view, and Move, Rotate, Push/Pull, Delete and the keys leave it as it is - so you can look all round the house without moving anything by mistake. Click the lock again to unlock.

Reference picture

 Ref. picture (MEASURE AND VIEW on the left) puts a photo or drawing of the house you want on a card over the 3D view, to compare as you model. Drag its top bar to move it, the corner at its bottom right to size it. The magnifiers zoom the picture - or turn the mouse wheel over the picture (the house does not zoom then); drag a zoomed picture to look round it. The half circle makes it see-through (click again for less), x closes it. It is kept with the project.

Split along a line (S)

Split: click as many points as you like to draw the cut line, then Enter (or click the last point again). Draw it on the ground or a top face and the cut goes straight down along your line, through every wall and solid it crosses. Draw it on the side of a wall or a shape and the line stays on that face (even past its edges) and the cut follows your line through that face - a curve drawn on a wall cuts the wall along that curve, following every dot. The first dot can be just past an edge: the second dot decides which face you are drawing on. Draw all the way round a piece (back to the first dot) to cut that piece out, like a window shape out of a wall. Each piece can then be moved, deleted or painted on its own. Backspace takes back a point; Esc stops.

Split along dots

Split along dots: put dots on a wall or a shape with the Dots tool (an outline - a door frame, an arch, any shape - or an open line of dots across it), then press Split on the card that shows (or the Split tool, S): it is cut exactly along your dots, and a closed outline comes out as a piece of its own.

Join walls

 Join walls (CHANGE on the left): click a wall, then the wall it should meet. Their ends run on (or are cut back) to meet in a clean corner where their lines cross; a wall that meets the middle of another makes a T. Windows and doors stay where they are.

Copy as it is

 Copy as is (CHANGE on the left): a wall with its windows and doors onto other walls - click the wall to copy, then each wall that should be like it (its own windows and doors are replaced, measured from the wall's middle); or everything in a room into other rooms - click inside the room to copy, then inside each room it goes to (furniture, lights and fittings in the same places from the room's name label).

Corner glass

 Corner glass (MODERN): every corner where two walls meet shows a blue dot. Click the corners you want - they turn orange; click one again to leave it out, A takes them all - then Enter and pick the glass. Each corner gets glass on both walls running into the corner, with no post.

Rescue a project that will not open

If a project does not open, Nestify offers Rescue (it is also in This house, and in Find a tool): it reads what it can - even a file cut off while saving - puts right what it can (walls with missing ends, missing sizes, doubled corners), hides anything that still cannot be built and marks it "(needs fixing)", and opens the house with a list of what was wrong: Show takes you to each one, Delete removes it. Save keeps the rescued house as a new file; the damaged one is not changed. A piece that cannot be built never stops a house from opening: it shows as a small red box until it is fixed.

Wall build-ups (what a wall is made of)

Click a wall, then Build-up (on the right, or What the wall is made of on the 2D plan's wall card): 220 double brick, 220 face brick, a 280 cavity wall, an insulated cavity, 140 or 190 block, a single brick inside wall, a drywall partition, a timber frame, a concrete wall, or Your own build-up (layers from the outside in, each a material and a thickness, saved for next time). Then pick which walls: this one, every outside or inside wall on the floor, or every wall. The wall's thickness becomes the sum of its layers. The plans draw each layer in its colour, the sections show the layers, Quantities and cost counts each one (bricks per leaf, blocks, wall ties across cavities, insulation, boards, plaster on the plastered faces) and the IFC file carries them. Typing a new thickness by hand makes it a plain wall again.

Boxed eaves

Eaves (roof card, or a drawn roof's Settings): open, or boxed: a fascia board along every eave and a soffit back to the walls, in white (225 or 300 mm), the wall colour or charcoal; the white band under the roof of many South African houses.

Files for CAD and building programs (IFC, DXF)

Export > Building model - IFC saves the house in the open format architects' and engineers' building programs read: every storey, the walls with their door and window openings, the doors and windows, the floor slabs, the rooms with their names, the roof, stairs, railings, furniture and the rest, each as what it is, with colours, materials and the layers of walls that have a build-up. Export > Elevations and sections for CAD programs - DXF saves the four elevations and two sections as lines (hidden lines left out), with ground line, floor levels and titles; the 2D plan DXF was already there. Open (or Import a plan) also opens a DXF floor plan from a CAD program: walls drawn as two lines become walls with their thickness, a door swing makes a door, window glass drawn in a wall makes a window, other gaps stay open. Layers named WALL are used when the drawing has them. The window that opens it shows what was found and lets you set the drawing's units and the wall height before Build the walls.

Draw the plan in 2D

The pencil-and-ruler button (top bar, and on the left rail) opens the **2D plan board**: a flat drawing board. The plan is **its own drawing**, saved with the project but separate from the 3D house: what you

built in 3D does not show on it, and drawing on it does not change the 3D house. When the plan is ready, **Build 3D** (top) puts that floor's walls, doors, windows, room names and furniture into the 3D house (boxes and shapes built in 3D stay; Ctrl+Z in 3D undoes it). **Start from the 3D walls** copies the 3D house's walls into an empty plan. "+ Floor" adds a floor to the plan.

- **Tools** (left): Select / move (L, D, W, O ... keys too), **Wall** (click corner after corner, type a length + Enter for an exact wall, click the first corner to close a room, Esc or right-click ends; it snaps to corners, to walls and to straight lines), **Door** and **Window** (click on a wall; choose the kind on the right), **Opening**, **Room name**, **Measure**, **Text**, **Furniture** (the library; the piece lands in the middle of the view, drag it into place), **Erase** and the **Hand**.
- **Select**: drag furniture, slide doors and windows along their wall, drag a wall, or drag a wall's round end to move that corner (the walls joined to it follow). The panel on the right changes a wall's length, thickness and height, a door or window's width, turns furniture, names rooms. Delete removes what is selected; Ctrl+Z undoes.
- **Looks** (top): **Colour** (red walls, tiles, furniture from above), **Technical** (black walls, dimension chains, sizes), **Blueprint** and **Sketch**. Switch any time; Nestify remembers your choice.
- **Save**: this view as a picture in the look you chose, the colour plan, or the A3 plans to print.
- The mouse wheel zooms where the mouse is; right-drag or middle-drag slides the paper; Fit shows the whole floor.

Doors and windows: select one and drag its round ends to make it wider or narrower, or type its width, height and (windows) sill on the right, use - 100 / + 100 mm, or pick a standard size. **Split (S)** cuts a wall in two where you click (each half can then be moved or changed), or cuts a railing in two.

Fixing railings, roofs and pergolas: select one and its corners show as round dots: drag a dot to move that corner, click a small + in the middle of a side to add a corner, click a corner and press Delete to remove it. **Split** on a roof (or pergola) cuts it in two with two clicks (where the cut starts and ends); select a piece and Delete removes that part. **Review** (top bar) shows a quick 3D sketch of the plan and a list of checks (walls whose ends do not meet, rooms without names, missing doors or windows, doors taller than their walls, roofs not over walls); Build 3D shows the same review first, with **Back to fix**.

Roofs on the 2D plan

Pick Roof on the plan's left and click its corners, or on the roof tool's card press Roof over all the walls. The card on the right sets the style, pitch, overhang, covering and colour. Then choose where to finish it: Finish it here on the plan keeps it as it is (Build 3D shows it later), or Design it in 3D Studio builds the house in 3D and opens the roof in the Nestify Studio for dormers, cut-outs, glass panels and any shape.

Drawing walls: thickness, faces and outlines

With the Wall tool (3D) or Wall (2D plan) pick the thickness on the card: outside 230, inside 110, thin 150 or your own. **[** and **]** make the walls you are drawing 10 mm thinner or thicker as you go, and Backspace takes back the last wall. "The line you click is" chooses what your clicks mean: the wall's **Middle**, the **Inside faces** (the walls go outwards, so the room is exactly the size you click) or the

Outside faces (the walls go inwards, so the house is the size you click). Clicking the first corner again closes the room cleanly.

Outlines shows the walls as their two faces only, in 2D and in 3D. In the 2D plan, with Select, drag a wall's face: the wall gets thicker or thinner from that side while the other face stays, and the walls joined to it follow. Drag the middle of a wall to move the whole wall; the walls joined to it stretch to stay connected. In 3D, Push/Pull on a wall's side does the same. Measure all puts dimension lines along every wall and between its doors and windows.

Doors and windows by a box or by points

When you trace a plan, leave the walls open where the doors and windows are. Then press  **Fit door** (left, or Doors, windows on the Plan picture card) and choose how to show where it goes:

- **1 click** - click in the gap between two wall ends: it is filled.
- **2 points** - click where the door or window starts and where it ends.
- **4 corners** - click the four corners of the opening (on a wall's face they also give its height and sill).

Pick what goes there (door, wide, double, sliding, garage, pivot front door, window, tall, high, glass wall, or just an opening). It fills exactly that space and never gets wider than the wall. When the points hold the ends of two walls, the two walls become one wall with the door or window in it, so the roof, the rooms and Check and fix see one unbroken wall. Points past the end of a wall make the wall longer; points away from any wall make a short wall for it. Check and fix also finds every gap left between walls in line and can fill them all at once.

Import a plan

Press the plan icon (top left), or drop a file on the window. Pictures (JPG, PNG, BMP, TIFF, WebP) and PDFs (every page; use the arrows to pick the one with the floor plan) are accepted.

1. **Sheet** – in a photo the paper is found and flattened automatically (perspective removed, standard paper shape kept), then squared up. If the guess is off, go to step 1 and drag the 4 cyan corners. A scan or PDF page is used as it is.
2. **Walls** – every wall is found: solid, hatched or hollow (drawn as two lines), straight or slanted. The sheet frame, title block, dimension lines, text, furniture, stairs and door swings are ignored, and the walls are joined at corners and T-junctions. Drag an end to fix it (ends that meet move together), drag on empty paper to add a missing wall, tap a wall to delete or split it.
3. **Size** – tap one wall you know (the longest is best), type its real length and press Set (or drag the ruler over any known length). Set the wall height; thickness is measured from the drawing.
4. **Build 3D** – gaps in walls become doors and windows drawn in solid walls become windows.

Moving round the view

Drag turns the view. To **slide** it (the ground and the house move together, as if you dragged the paper): right-drag, Shift-drag, the arrow keys, or turn on the **hand** (H, or the hand button on the left

rail) and drag. The wheel zooms, F frames the house, T looks straight down.

Getting around: fly to, views and What's next

Double-click anything - a wall, the roof, a sofa, a spot on the ground - and the camera glides over to it and turns round it from there. **Views** (floor bar) shows the views you were in as small pictures: click one to go back; Alt + Left goes back one. The **What's next** card (left of the 3D view) offers the next step for where the house is: draw walls, close the rooms, doors and windows, the roof, furniture, then a photo, a walk-through or the cost. x hides it until the next stage.

Guides

Press **Guide** (left), click where a guide should go, then press **V** for a vertical guide (up and down the plan) or **H** for a horizontal one (across), or pick it on the small card that comes up. Click as many as you like; Esc stops. Clicked on a wall, the card also offers a guide along that wall at a distance you type. The Wall tool snaps to guides, and **No guides** removes them all.

Editing walls, doors and windows

Tool	Key	What it does
Select	Space	Click a wall, the floor or the roof. Drag a selected wall's **grip** to move that corner – walls joined there follow.
Wall	L	Click corner after corner. Snaps to **Endpoint**, **Midpoint**, **On wall**, **Intersection**, the **red** / **green axes**, **Parallel** / **Perpendicular** and **From point**. Type a length + Enter for an exact wall. Ending on a wall splits it, so the joint stays editable.
Move	M	Drag a wall; walls joined to it stretch (Ctrl pulls it free). Drag a door/window along its wall. Type a distance + Enter afterwards for an exact move.
Rotate	Q	Drag to turn the selected wall (snaps to 15°).
Push/ Pull	P	Drag any face: the top changes height, a side changes thickness (walls) or width/ depth (objects), a wall end changes length. The face lights up orange; the opposite face stays put (Ctrl: both sides). Type a size + Enter, or + / – for 5 cm steps.
Split	S	Click a wall where it should become two.
Opening	D	Drag along a wall, then pick a door, window, passage or separate wall.

Click the orange arrow strip on the Layers or Properties panel (or press `[` / `]`) to fold it away for a bigger 3D view; on a small window they fold away by themselves. T toggles top view, F frames the house, Delete removes, Esc cancels, Ctrl+Z / Ctrl+Y undo / redo, Ctrl+S save, Ctrl+O open. The **Measurements** box (bottom right) accepts `3.5`, `3500mm`, `350cm`, `12ft`, `30in`.

Keyboard resizing of the selection: Ctrl+←/→ thinner/thicker (narrower/wider), Ctrl+↑/↓ lower/higher, Alt+←/→ shorter/longer (shallower/deeper), Shift for 1 cm steps. A tips card in the 3D view (lightbulb button) always lists the keys for what is selected.

Rotate with the protractor

Select an object and press Rotate (Q): a see-through glass protractor with the degrees on it appears round it. Point at the face you want to turn about: on the top the circle lies flat and the object turns round; on a side it stands up and the object tips forwards or backwards; on the front it faces you and the object rolls. Drag round the circle - it clicks to every 15 degrees, and the angle shows while you drag. Walls still turn about their middle, and -15 / +15 step them.

Drawing shapes

- **Rectangle (R)** and **Circle (C)**: draw on the ground or on the top face of anything, then move the mouse up and click to pull the shape into a solid (or type `3,2` / a radius, Enter, then a height, Enter). Draw on top of a solid to stack another one. Solids are objects: Move, Rotate, Push/Pull, copy, delete.
- **Line (N)**: click corner after corner for any outline, **A** makes the next side an arc (click its end, move to bend it, click); close on the first corner or Enter, then pull it up into a solid.
- **Paint bucket (B)**: choose a material or a colour, click walls, roof, floors, rooms or solids.
- **Groups**: Ctrl+click to select several things, **Ctrl+G** groups them (move, turn, copy and delete together), **Ctrl+Shift+G** ungroups.
- **Scenes**: save the current view (camera, style, floor) and glide back to it.
- **Tags**: show or hide whole kinds of things (Furniture, Electrical, Plumbing, Garden, Shapes...).

Shape push/pull

The Shape pull tool (U): pick on its card how to draw the outline on a face - Rectangle, Circle, Polygon, Dots (corner after corner), Curve, or **Auto**. Auto outlines the whole face you click and works out how far it must be pulled to meet the next wall or object: **Fill to it** (or Enter) does it - the end of a wall pulled across a gap becomes a longer wall that meets the next one. Edit dots resizes any outline. It works on walls, the floor slab, the roof and any object, furniture too.

Boxes as houses: doors, windows, split, paint one face, walk through

Many people build with the Rectangle tool: draw, pull up into a box, and use the boxes as walls.

- **Door, window, passage** (left rail): drag across the side of a box where the opening goes. The card offers the usual doors and windows (or a plain passage); the hole is cut right through and a door leaf or pane of glass is put in it.
-  **Split** cuts a box in two where you click, so each part can be moved, painted or pulled on its own.
-  **Paint bucket** on a box paints only the face you click; Shift + click paints the whole box. To paint part of a face, draw round it with Dots and choose **Paint area**.

- **Walk** works in houses made of boxes too: their sides are walls you cannot walk through, and their doors stand open so you can walk in.
- **A box inside a bigger one**: click again at the same place to pick the next thing behind (each click goes one deeper, then back to the front). Clicking through an opening picks what is behind it.

Dots push/pull (shape anything)

The Dots tool (left rail, or **U**) shapes the solids you draw: draw on a face, then push or pull it. Drawn on the top of a wall and pulled up, the outline becomes a new shape of its own standing on the wall (a higher wall, a parapet, a pier): it is selected straight away, and Push/Pull, Move, Paint and the rest work on it alone. After closing an outline you can also pick Push/Pull (P) and drag: the outline waits for it. The Rectangle, Circle and Line tools draw on the tops of walls too, so a rectangle drawn on a wall's top and pulled up is a new block on the wall.

1. Pick the Dots tool. A card at the bottom of the 3D view asks how to join the dots: **Straight lines** or **Smooth curve** (Nestify remembers your choice).
2. Click a face of a shape (a box from Rectangle, a cylinder from Circle, a Line outline): that is dot 1.
3. Click more dots around the outline you want. They snap to corners and edges. Backspace takes one back.
4. Click the first dot again (or double-click, or Enter). The outline turns **yellow** and a card asks what next: * **Edit dots** (or **E**): drag any dot to a new place, click on a line to add a dot there, Backspace removes the last one. **Done** (or Enter) when the outline looks right. * **Paint area**: asks what to paint with (a colour, water, tiles, grass ...) and paints only the inside of the outline, for example a darker panel on a wall or a doormat on a floor. * **Make a pool** (on a top face): digs the outline 1.4 m into the shape and fills it with clear water just below the top: a swimming pool or a pond of any shape. Select "Pool water" to change its level or colour. * **Push / Pull** (or Enter, or just right-drag): the outline turns **green**, ready to push or pull.
5. Hold the **right mouse button** and drag: **into** the face pushes the outline in (a recess, a window bay, a hole right through if you go far enough), **out** of the face pulls it out (a bump, a bay, a balcony box). The shape changes as you drag; let go to keep it. Looking straight at a face, drag up to pull and down to push. Or type a distance + Enter (negative pushes in).

If your dots go round (nearly) the whole face, the whole face moves, like Push/Pull. Every new face can be shaped again, so a modern house can start as one big box: push in the bays, pull out the overhangs. Ctrl+Z undoes a step. Esc cancels.

Shape looks: solid, glass, frosted, glowing

Select a shape you drew (Rectangle, Circle, Line, or one edited with Dots). Under **Look** on the right: **Solid**, **Glass** (clear, see-through), **Frosted** (milky, blurry glass that lets light through) or **Glow** (lit from inside; in the evening it lights the walls and ground around it, and in a photo render it is a real light). **Colour** changes the tint of the glass or the colour of the glow. Material still offers brick, wood, tiles, your own pictures and the rest.

Settings for anything you select: see-through, glow, light

Select something (Ctrl+click several) and press **Settings** in the panel on the right: * **See-through**: 0 to 90 %. * **Look**: Solid, Frosted or Glow. Any object can glow now, not only shapes. * **Light** (for lamps and glowing things): brightness 0 to 300 %, the colour of the light (warm white, white, cool white, red, orange, yellow, green, blue, purple, pink), and which way it shines: all round, down, up, a spot straight down, or aimed (turn and tilt), with the beam width. The light shows in the evening (Sun card) and in photo-real pictures.

New in the library: **Swimming pool** (tiled, coping, a step, clear rippling water), **Pond**, **Frame (square)** and **Frame (flat ring)** that you can size and make glow, and an **LED strip**. Any shape can be water: Material → Water, or the paint bucket's Water.

The tool rail (left)

The tools sit in groups: **Select and draw** (select, wall, line, arc, rectangle, circle, polygon, Dots, opening), **Change** (move, rotate, push/pull, scale, split, offset, Sweep, array, mirror, paint, group, delete), **Solid tools** (union, subtract, intersect, trim), **Build** (all the Build tools, railing, roof, structure, ceiling, 3D text, add a part, library, import a model), **Measure and view** (tape, measurements on/off, section, hand, zoom, top view, frame, walk, hide, show everything) and **Scene** (site, sun, sky, tiles and stone, the 2D plan). Hover over a button to see what it does.

Wider or narrower: drag the thin strip on the rail's right edge for 1 to 4 buttons across (3 by default, so there is little scrolling). The Layers panel and the right panel can be dragged wider or narrower by their edge strip too (a click on the strip still folds the panel away). Nestify remembers the sizes.

Solid tools: select one shape you drew, Ctrl+click a second one, then Union (one shape), Subtract (the second cut out of the first: a hole, a niche, a pool), Intersect (only where they overlap) or Trim (the first cut out of the second, both kept).

Smart helpers

- **Furnish** (Build group, the house-and-heart button): every named room gets fitting furniture with its back against a wall and clear of the doors: beds and wardrobes in bedrooms, sofa, TV and coffee table in the lounge, a dining table, kitchen counter, stove and fridge, toilet, basin, bath and shower, desk and bookshelf in a study, washer and geyser in the laundry, a car in the garage, and a ceiling light in each room. Rooms are recognised by their names (Lounge, Kitchen, Bedroom ...). Furnish again rearranges only what it put in; "Take out what Furnish put in" removes it. Things you placed yourself are never moved.
- **Running cost:** the status bar shows the building cost of the house as it is now, updated a moment after every change, with how much the last change added or saved. Click it for the full bill of quantities.
- **Describe the picture** (Scene group, the sparkles button): type something like "modern house, golden hour, warm lights, few clouds, winter" and Nestify sets the sun time and month, the sky, the lamps and (for modern, contemporary, farmhouse, tuscan, face brick, stone or wooden) the wall, roof and frame finishes, then offers the photo-real picture.
- **Guided walk video** (Scene group, or Export): click where the walk starts (the front door, or

outside it), then click inside each room in the order to visit, and press Enter. Nestify finds the way through the doors by itself, walks at walking pace with the doors open, and stops in every room to look round it (its name and size on screen) before going on to the next: an MP4 in the size you choose.

- **AR page for phones** (Scene group, or Export): one HTML file with the house inside. Send it to a phone, open it in Chrome (Android) or Safari (iPhone) and press "See it in your space": the house stands on a table (1:50) or full size on the real plot. The phone needs to be online.
- **Door or window anywhere** (Select and draw group, the window button): pick a door or window, then click on a drawn wall or on the side of a box: it is cut in right there. In a box it gets a frame (jambs, head and, for a window, a sill) round the door leaf or glass, all grouped together. Click again for more; Esc stops.

Build tools: railings, roofs, ceilings, Sweep, 3D text

The toolbox button on the left (and the railing, roof, ceiling and sky buttons under it) holds the tools big modelling programs have. Everything they make is an ordinary object: move it, turn it, copy it, paint it, and change every setting later with **Settings** (right panel).

- **Railing**: pick a type (glass panels, vertical bars, horizontal rails, steel cables, timber slats, wrought iron, picket fence or a solid panel), then click point after point where it goes; Enter finishes, clicking the first point takes it all the way round. Settings: height, post spacing, post size, gap between bars, number of rails / cables, handrail (round, square, flat cap, none), bottom rail, square or round posts or frameless glass, and the colours of the frame and of the bars / glass / slats.
- **Roof of your choice**: pick hip, gable, shed (lean-to), flat, gambrel (barn), mansard or pyramid and click the corners of what it covers: a garage, a patio, a carport. Drawn on the ground, it sits on top of the walls. Settings: type, pitch, overhang, thickness and covering (tiles, metal sheets, slate, concrete, timber, glass / polycarbonate). Make as many roofs as you like. The house's own roof (which follows every wing of the plan) is still there too.
- **Ceilings**: flat, bulkhead (dropped edge), tray (stepped up), LED cove (with a glowing strip in its colour), flat with cornice, coffered (a grid of beams) or sloped; in every room at once, in the selected room, or over any outline you draw.
- **Sweep**: a profile swept along a path: round pipe, square bar, rectangle, skirting / moulding, kerb or gutter, any width and height, open or as a closed loop.
- **3D text**: words standing up or lying flat, bold or regular, any height and thickness; make them glow in Settings for a sign.
- **Array copies**: many copies in a row (step across, along and up, like 1.2, 0) or round a circle (p 45 for a copy every 45°).
- **Mirror**: flip what is selected left to right or front to back.
- **Offset outline**: a new outline inside (0.1) or outside (+0.1) the top of a shape you drew, ready to Push/Pull: a border, a step, a kerb.

Railings, roofs and pergolas can also be drawn on the **2D plan** (Railing, Roof and Pergola tools): the plan, the colour plan and the printed plans show railings with their posts, and roofs dashed with

their ridges, hips and valleys (ceilings show their bulkhead line).

Roof structure and pergolas

Structure (Build group): timber or steel **trusses** with battens / purlins, or **rafters** with battens, under a roof you drew, or under the house's own roof; and **pergola** / **carport** frames (timber or steel posts, beams and joists, open or covered with slats, polycarbonate, metal sheets or shade cloth). In a roof's Settings choose the structure, its spacing, and **Hide the covering** to show the structure. To see the structure under the house's roof, hide the roof (the eye next to Roof in Layers).

Tiles and stone

Material > Tiles and stone (or the Tiles button in the Scene group) on a room's floor, a wall, a shape or the roof: white, grey and black marble, slate, patterned terracotta, white and pool-blue mosaic, crazy paving in sandstone or blue-grey, hexagons, subway tiles, herringbone wood, planks, stacked stone cladding, terrazzo and a black-and-white checker. Each comes with its real size, so the tiles are the right size at once.

Sky

Sky (the cloud button on the left): clear blue, a few clouds, cloudy, overcast, sunset glow or dusk, or **your own sky picture**. A 360 panorama (twice as wide as high) fits best; an ordinary photo of the sky is mirrored round the house so it joins up. The sky shows in the Real look and in reflections.

Shadows and see-through for anything

Settings has **Casts a shadow** / **No shadow** for anything you select, including the lines and outlines you draw (a Line outline is a shape too: give it glow, see-through or no shadow like any object).

Auto adjust

Ctrl+click the things to tidy (boxes, walls made of boxes, furniture) and press **Auto adjust** in the panel on the right. Within 12 cm it squares them up with the biggest one, puts things that hover or sink just a little right on top of what is under them (or on the ground), closes small gaps and overlaps so faces touch, and lines up edges that are nearly in line. The biggest thing stays where it is; things further apart are left alone. Ctrl+Z puts them back.

Edit group (build one thing on its own)

Open a group to work inside it: double-click a shape, a piece of furniture or a group, or select it and press **Edit group** on the right. The rest of the house turns into faint ghost lines and the camera moves in. Draw rectangles, circles and lines, push / pull them, or add Library items: everything you make

joins the group, and only the group's pieces can be clicked and dragged (even behind walls). **Done** (top left) or **Esc** brings the whole house back, at the view you had before.

Your own pictures as materials

Any picture can be a material: floor tiles, marble, wood, brick, stone, wallpaper, a logo. **Material > Your own picture** on a room floor, a wall, the roof or a shape you drew: choose the picture and say how big one picture is in real life. Nestify makes it repeat without seams; it shows in the Real style and in photo-real pictures. A house keeps up to 9 pictures, and they can be reused anywhere.

Which picture works: one tile per picture. Crop the photo to exactly one tile (or one plank, one brick), edge to edge, taken straight on in even light, and type that tile's size (0.6 for a 600 mm tile). A picture of several tiles also works if you type the size the whole picture covers (2 x 2 tiles of 300 mm: 0.6). A photo of a whole room or of tiles at an angle does not work: it repeats as a pattern of photos. Nestify shows this guide before you choose, and warns when a picture looks like many tiles.

Shapes can also be made of **glass** (Material > Glass) - for your own windows, glass boxes and balustrades.

Floors, stairs and rooms

The bar at the top left of the 3D view lists the floors. **+ Floor** adds a storey on top (empty, with the outside walls copied, or a full copy). Click a floor to edit it; the floors above it are hidden so you can work. **All floors** shows the whole house, the roof always sits on the top floor, and ⋮ renames or deletes a floor. Put **Stairs** (Library → Furniture) on a floor and the floor above gets a hole over them; when walking you climb them, or press **PageUp** / **PageDown** to change floor.

Rooms are found automatically from the walls (a doorway counts as closed). Each room shows its name and area in the view (the **Rooms** chip turns the labels off) and in the Layers list, with the total area at the top. Click inside a room to rename it and give it its own floor material and colour; names stay with the room when you move walls.

Walk through the house

Press the walking-person button (or **W**). You start outside the front door at eye height (1.6 m) and walk in like a visitor: **W**/**S** or ↑/↓ walk, **A**/**D** step sideways, ←/→ or **Q**/**E** turn, drag to look around, **Shift** walks faster, the wheel takes a step, and clicking the floor walks you there. Walls, windows and furniture stop you; doors swing open and passages let you through. A mini-map shows where you are (click it to go there) and a **Rooms** list jumps into each room (found automatically, with its area). **Esc** stops walking and puts the camera back where it was.

Structure parts, and lifting objects

The library has a **Structure** tab: round and square concrete columns, brick piers, steel H-columns, timber poles, concrete / steel / timber beams, lintels, roof trusses, purlins, strip footings, slabs, boundary walls, palisade fencing, a carport, a water tank, downpipes and solar panels. Resize

any of them like other objects.

To lift anything off the floor (a beam, a shelf, a solar panel onto the roof): **Page Up** / **Page Down** (Shift for 1 cm steps), the **Up 10 cm** / **Down 10 cm** buttons, or **Shift + drag** it up and down. Footings start below the ground; the X-ray view shows them.

Library packs (add models, materials and ready houses)

A pack is a .zip of 3D models, materials and ready houses that anyone can make and share. Open **Packs** (on the left, or the Packs button in the Library) and press **Add a pack**: its models appear in the Library under the pack's own name, and its materials and ready houses are on the Packs card (Materials puts one on the selected wall, every wall, a floor, the roof or the ground; Houses opens one to start from). **Remove** takes the pack away again; copies already placed in your houses stay. Adding a newer copy of a pack replaces the old one.

Packs are only pictures, models and a description file: nothing in a pack is ever run, and files that try to point outside the pack are ignored.

Make a pack shares your own things: give it a name, choose your models (My items and Downloaded), this house as a ready house, or both, and save the .zip. Anyone with Nestify adds it with Add a pack.

Models downloaded from the Online library are kept together in the Library's **Downloaded** group, and are reused when you add the same one again.

To make a pack by hand, put `pack.json` at the top of a folder and zip it:

```
json { "name": "Modern cars", "version": "1.0", "author": "Your name",
"description": "Three cars for the driveway", "items": [{"name": "Sports car",
"file": "models/sports_car.glb", "size_mm": [4500, 1900, 1300], "icon": "car"}],
"materials": [{"name": "Oak floor", "file": "materials/oak.jpg", "tile_mm": 1200}],
"houses": [{"name": "Beach house", "file": "houses/beach.nestify"}] }
```

Models can be .glb, .gltf, .obj, .fbx, .dae, .stl or .ply; size_mm (width, depth, height) and icon are optional.

Library: furniture, electronics, cars, trees, flowers

The sofa button opens the Library: 37 items (sofas, beds, dining set, wardrobe, kitchen, bathroom, TV, fridge, washing machine, stove, computer, air conditioner, cars, SUV, pickup, trees, pine, palm, bushes, flowers, hedge, fence, bench, pool). Click one, move the mouse to place it and click to drop it. Every object can be dragged (Move), turned (Rotate), resized (Push/Pull or the numbers on the right), recoloured, renamed, copied (Ctrl+D) and saved to **My items** with its name, size and colour. **Import .obj model** adds your own 3D models. My items live in `~/.sketchit/library.json` and `~/.sketchit/models/`.

Online library: free models and materials, and models to buy

The **Online** button (left) opens free 3D models and materials (CC0, free for any use). Before

anything is downloaded, your computer's own Save window asks where to keep it: pick any folder, make a new folder (the New folder button in that window) and give it a name - so downloads for different houses or rooms are not all mixed in one place. Nestify opens that window in the folder you used last time (at first Documents > Nestify downloads). A model is then saved there as one .zip with its pictures (Import a 3D model opens it again), and Nestify asks its name in your Library and which Library group to keep it in: one of the groups, or a new group you type (for example Bedroom 2 or Garden ideas), which then shows in the Library like the others. A material is saved there as its picture (.jpg). A model you downloaded before asks whether to use the one in your Library or save another copy. On the left it suggests what each room of your plan needs (a sofa and a TV for the living room, a bed and side tables for a bedroom ...), or type what you want and press Search: plurals and everyday words work ("cars", "couch", "fridge"), and things already in Nestify's own Library show first, marked "In your library". Click a picture: a model goes into its room, a material onto the selected wall, floor, roof or the ground. Downloaded models stay in the Library group you picked and are reused next time.

Models to buy. The free library does not have everything (it has very few cars, for example). Under the results, **Buy** opens the same search in a few model shops in your browser. Buy and download the model there (choose .glb, .fbx or .obj), then press **Add a bought one**: it comes into the house and is kept in the Library's **Bought** group. Payment happens on the shop's own site; Nestify takes no part in the sale. Bought models are left out of packs you make, because shop licences rarely allow sharing them. You can also drag a model file (or a pack) straight onto the Nestify window.

Lights

The library's **Lights** section has ceiling, pendant, wall and garden fittings: Ring LED (circle), Double ring LED, Round ceiling light, Square LED panel, Pendant (glass globe), Pendant (dome shade), Hanging filament bulb, Chandelier, Track spotlights, square Downlight, Wall light (up and down), Garden bollard, Garden spike spotlight and String lights. Ceiling ones hang from the ceiling of the floor you are on. Settings sets each one's brightness, colour and beam.

Your own 3D models

Library → Import 3D model (or **Build tools → Import a 3D model**) reads **.obj**, **.glb** / **.gltf**, **.dae** (Collada) and **.stl**. Each model keeps its parts: one per material, with its colour (and the average colour of its textures). Each part is recognised from its name and colour as leaves, trunk / wood, glass (see-through), a light (it glows and lights the room) or the body; a model with leaves is filed as a plant. Settings shows every part with its own colour button, plus see-through, glow and light like any other object. Very detailed models (over 60 000 triangles) are simplified on the way in. Compressed glTF (Draco) cannot be read: export it again without compression.

Stairs, railings, site, electrical and plumbing

- **Stairs** (Library > Building): straight, **L-shaped** and **U-shaped** with a landing and a sloping handrail; the floor above gets a stairwell and the walker climbs them, landing included. Also a steel balcony **railing** and a **glass balustrade**.

- **Sloping site**: the mountain button opens the Site card (slope %, the direction it falls, the level platform around the house). The ground is drawn sloping with shadows; **driveway, brick paving, stone path, gravel, instant lawn, timber deck and retaining wall** (Library > Garden) follow it, cars and trees stand on it, and the walker walks on it. The estimate adds paving areas and the cut-and-fill to level the platform.
- **Electrical and plumbing points** (Library > Electrical / Plumbing): plug sockets, switches, TV/data points, outside lights, distribution board and stove isolator snap to the nearest wall at the usual height and face the room; ceiling lights and downlights hang from the ceiling; taps, garden taps, shower, geyser and floor drains. They show as plan symbols (layers ELECTRICAL / PLUMBING in the DXF) and the estimate counts each one plus wiring per electrical point and pipework per wet point.
- **Sun study**: the Sun card films shadows from sunrise to sunset (a frame every 20 minutes, time stamped) as an animated GIF, or MP4 when `imageio[ffmpeg]` is installed.

Ground

The Site card (the hills icon on the left) has a Ground button: grass, dusty soil, gravel and stones, concrete, paving bricks, sand, tar, or **your own picture**. A photo of a lawn, a driveway or a farm is evened out and made seamless, so it repeats without joins; set how wide the area in the photo is. The ground shows in the 3D view, on a sloping plot and in photo-real pictures.

Hills and dips

The Hills button (left, or Hills and dips on the Site card): click the ground to put a dot, and it rises into a round hill (press Dip first and it sinks). Click a dot to pick it: High and Wide on the card change it while you watch, Move puts it somewhere else, Delete and Clear take dots away. Several dots make a ridge, a valley or a rolling garden. The house keeps its level platform, and trees, paving and cars settle on the new ground.

Modern glass and solar panels

Windows: **Glass wall** (floor to ceiling), **Slim frame** and **Clerestory strip**; doors: **Glass slider** and **Glass stacker**. They have slim black aluminium frames split into tall panes, like modern houses. Change the frame colour of any wall's windows and doors (black, charcoal, bronze, silver, white) with **Material > Window and door frame colour** - for one wall or every wall. With a **flat** roof and a deep overhang (Roof settings) and **Natural stone cladding** from the finishes catalogue you get the modern villa look.

Solar panels (Library > Structure) lie on the roof: move one over the roof and it settles on the surface, tilted with the slope and facing down it. Ctrl+D copies it for the next panel.

Real look

The default **Real** style paints plaster, brick, stone or timber walls; wood, tile or concrete floors; tiled, metal or slate roofs; wooden doors and glass windows, under sky light on grass. Pick Material and

Colour on the right for a wall, the floor or the roof. Night, Sketch, Neon wire and Blueprint are still there.

Live style pictures

The style pictures on the right turn slowly round the house while you are resting and show exactly your view (with your latest change) while you work. Slow computers get fewer updates; Settings can turn them off.

Pencil and Glass styles

Two more styles on the right: **Pencil** draws the house like a hand sketch - white paper, graphite edges and hatching on the shaded sides. **Glass** makes every face see-through with all its edges showing, so you can see how pieces fit together while you build (it keeps depth, unlike X-ray).

Roof that follows the plan

The roof is built from the real footprint: an L, T or U house gets one wing per arm, the ridges run into each other and meet in valleys, and courtyards stay open. Hip, gable, shed, flat, gambrel and mansard all work this way; slanted outlines get a true hip over the outline. Eaves overhang is measured from the outside face of the walls. While the Wall tool is active the roof steps aside so you can see the plan.

Sun and shadows

The sun button on the left opens the Sun card: time of day, month, place (city list), and which way north points on the plan. The house casts real shadows on the ground, on itself and from trees and cars, and the light turns warm in the morning and evening. **Shadows** and **Real sun** can be switched off.

X-ray (engineering view)

The X-ray card in Styles turns the house into a see-through engineering model: glowing blue surfaces, every edge drawn (hidden ones too) and a mesh of lines across walls, floors and furniture. Click any wall, object or anything you built and it glows blue while everything else turns into grey ghost lines, so you can study that one part inside the whole house. Click empty space to see everything in blue again.

Photo-real pictures

Press the camera-shutter icon (top right) or Export → Photo-real picture. Nestify hands the view you see to a physically based path tracer (Mitsuba 3): a real sun and sky for your place, day and hour; light that bounces around the rooms and comes in through the windows; glass that reflects; plaster,

face brick, stone, timber floors, tiles and roof tiles as real materials. Choose:

- **Size**: Small, HD (1280 × 720), Full HD, Square.
- **Quality**: Quick (a draft in seconds), Good, Best (for printing).
- **Light**: as the Sun card, Morning, Golden hour, Evening with the lights on, Night.

Photos need a processor with AVX instructions (almost every Core i3/i5/i7 or Ryzen PC; not low-power Celeron, Pentium Silver or Atom chips). On those PCs the Photo button says so and offers Picture of this view.

The picture sharpens pass by pass while you watch, is cleaned of grain, toned like a photo, and kept in Pictures/Nestify as well as wherever you save it. Walk inside (W) for a photo of a room: from inside, the windows are open to the sun and the sky.

Lamps (ceiling lights, downlights, floor lamps, outside wall lights) glow, and in the evening (or with Lights on in the Sun card) they light their own rooms in the 3D view and are real lights in photos. **Curtains** (open, closed, and net curtains that let light through) are in the library under Furniture; generated houses get curtains on their windows. **TVs** (on a stand, on the wall, a computer screen) can show a picture or play an animated GIF: select the TV and press "Picture / GIF on the screen". It plays in the 3D view and shows in photos.

Building rules check

The clipboard button in the top bar checks the house against SANS 10400 rules of thumb: habitable rooms at least 6 m² and 2 m wide, ceilings 2.4 m (2.1 m for bathrooms and passages), windows about 10% of the floor for daylight and 5% opening for fresh air, doors 750 mm, stair risers up to 200 mm and treads 250 mm, outside walls 140 mm or more, stairs to every upper floor. Tap a finding to go to it. A first check, not an approval.

Quantities and cost

The calculator button in the top bar opens a bill of quantities worked out from the model: strip footings (600 x 230) and floor slabs in concrete, bricks (52 per m² per leaf, +5%), cement and sand for mortar and plaster, paint (2 coats), floor tiles / wood / screed per room, ceiling boards, roof tiles or sheets from the real roof surface, roof timber, gutters, and every door and window. Each line is priced; labour and contingency are added as percentages to get the total and a rate per m². **Prices** edits every rate (saved in `~/.sketchit/prices.json`). The currency follows the computer's country (Windows region setting): rand in South Africa, naira in Nigeria, pounds in the UK and so on, with the built-in prices converted at today's exchange rates as a starting point. Rates are fetched in the background from open.er-api.com (free, no account) when the app starts, kept in `~/.sketchit/rates.json` and refreshed every 12 hours; without internet the last rates are used (**Update rates** in Prices fetches them now); pick another currency at the top of Prices, either converting the numbers or only changing the symbol; **PDF** and **CSV spreadsheet** save the estimate with a room schedule. Rules of thumb for planning, not a quantity surveyor's bill.

Tape measure

E (or the tape button) measures between two clicked points, snapping like the Wall tool. Kept sizes show in the 3D view and print on the 2D plan; Ctrl+Z removes the last, the floor menu clears them.

Plans, council pack and pictures

Export (toolbar) offers:

- **2D plan PDF**: one A3 page per floor at a standard scale (1:50, 1:100 ...), with walls, doors and their swing, windows, stairs, furniture, room names and areas, dimension chains, scale bar, north arrow and a title block. Sizes are in the unit chosen at the bottom right.
- **2D plan PNG** of the floor you are on.
- **Elevations and sections**: the plan PDF continues with the four elevations (named by the compass from the north in the Sun card) and sections A-A and B-B drawn from the 3D model: clean lines, pale-blue glass, timber doors, grey roof, cut walls / slabs / roof in solid black, level marks (FFL of every floor, wall plate, ridge) and the overall height. The section lines are marked on every floor plan. "Elevations and sections only" saves just those pages.
- **DXF** for CAD programs (layers WALLS, DOORS, WINDOWS, ROOMS, DIMS, FURNITURE; floors side by side, millimetres).
- **Picture of this view** at 2x or 3x the window size, without grips, labels or selection.
- The 3D model as OBJ or STL, as before (all floors and the roof).
- **Council submission pack** (PDF): site plan (plot boundary, building lines, street, roof outline, driveway, setback dimensions, coverage), every floor plan, elevations, sections, a 1:25 wall section (footing, DPC, slab, wall plate, roof build-up), and schedules (rooms and areas, doors, windows, checks, general notes). Every sheet carries the owner, erf and address from **Project details**, which the pack asks for the first time.
- **Share as a web page**: one .html file with the house in 3D inside it. Anyone can open it in a browser, on a phone too, with no app or internet: turn it with a finger or the mouse, Top / Front views, take the roof off, hide the upper floors, and see the room list.

Colour plan for clients

Export → **Colour plan for clients** draws the floor plan the way plan books and estate agents show it: thick red walls (or black, grey or blue), light tiled floors with small tiles in bathrooms, kitchen and laundry, paving in the garage and on patios, windows as pale blue glass, doors with their swing shaded timber, every piece of furniture, bathroom fitting and car seen from above in its own colours with a soft shadow, and each room's name in bold capitals with its area. Names move to a free spot so furniture never covers them. It is cut at 1.2 m like an architect's plan, so houses built from boxes work too. PNG for this floor, or a PDF with every floor.

Sides of the house (your own views in the plans)

Turn the 3D view to a side you like and press **Sides** (the bar over the 3D view) → **Save this view as a side**: Front, Back, Pool side ... **Plans to print** then prints your sides (four to an A3 page, named, not to scale) instead of the drawn elevations; the sections stay. Choose how they look on paper (Real,

Sketch, Pencil, Blueprint, Glass) in the same menu. Update, rename or delete a side there too. With no sides saved, the drawn elevations print as before.

Section view

The **Section** chip above the 3D view cuts the house open like a builder's section drawing. **Cut across** slices it at a height (1.2 m above the floor to start) so you look down into the rooms; **Front to back** and **Side to side** slice it upright, cutting away the half nearest you. Drag the slider to move the cut, and **Flip** to look from the other side. Cut walls show solid red, the way sections are drawn. Pictures you save keep the cut; **Off** (or the Section chip again) closes it.

Kitchen builder

Select the wall the kitchen goes against and press **Fit a kitchen on this wall** (on the right). Pick what you want: cupboards, a sink, a stove with an extractor hood and a fridge; without the fridge; with a tall pantry as well; or base cupboards only. Nestify uses the longest free stretch of the wall (doors and low windows are kept clear, up to 5.4 m), sizes the cupboards to fill it exactly, and leaves wall cupboards out under windows. The kitchen is one group: drag it to move it all, or Ungroup to change single units. The pieces are also in the **Kitchen** section of the Library.

Finishes catalogue

Select a room floor, a wall or the roof and press **Material**: at the top is **Choose a product**, a list of real finishes with their prices: floor tiles, porcelain, laminate, vinyl plank, solid wood, polished concrete, carpet tiles; paint colours, face brick, stone and timber cladding; concrete and clay roof tiles, IBR, corrugated and Klip-lok sheeting, slate. The 3D view takes the product's look, and **Quantities and cost** prices it on its own line (with waste). Prices follow your price list's currency. Add your own products in `~/.sketchit/materials.json`.

360 pictures and walkthrough videos

Export > 360 picture of a room: pick a room (or where you stand in walk mode) and Nestify saves a 360 photo from eye height. Post it on Facebook or open it on a phone and turn around in it.

Export > Walkthrough video makes an MP4: a slow flight around the house, a glide through your saved Scenes, then a look around every room with its name and size. Choose standard, HD or full HD.

Export the 3D model

Export > 3D model in colour: a **.GLB** (glTF) file opens in most 3D programs and 3D web viewers; **.OBJ** comes with a **.MTL** colour file beside it. Both include the roof, doors, windows and furniture, in metres. STL (for 3D printing) is still there.

Updates

Once a day Nestify checks the developer's website for a newer version and offers the download (Help > Updates checks now). Copies from the Microsoft Store are updated by the Store instead.

Files

Save writes a `.json` project (atomically). Unsaved work is autosaved every 45 s and offered back after a crash. Export writes the 2D plan (PDF, PNG, DXF), a picture of the view, or the 3D model as OBJ (most 3D programs) or STL (3D printing), in millimetres. Floors, rooms and the sun are saved in the project; older projects open as one floor.

Working with big files

The Layers list is updated in place instead of rebuilt (it used to take ~0.4 s per click), the roof is not rebuilt on every mouse move while dragging, and big changes (roof type, new plan, scaling, settings) are built in the background behind a "please wait" card.